

Here are the notes from the two videos, formatted for your Markdown editor.

Lecture Notes: Replicates and Sample Size

These two StatQuest videos build directly on each other. The first defines the types of replicates, and the second explains how they relate to the concept of "sample size."

Video 1: The Difference Between Technical and Biological Replicates

1. Technical Replicates

Definition: Performing the *same measurement* multiple times on the *exact same sample* (or biological source).

- **Example 1:** Take one blood sample from "some dude" and measure his gene expression three times from that single sample.
- **Example 2:** Take three blood samples from "some dude" *at the same time* and measure each one once. This is also a technical replicate because all samples originate from the same individual at one time point.

What do they tell you?

- **About the individual:** They give you a very accurate and precise measurement for that *one specific person*.
- **About the method:** They measure the **variability and accuracy of your measurement technique** (e.g., how much "noise" the machine or process introduces).

When to use them:

- When you only want to tell a story about a single individual.
- When you want to prove that your new measurement *method* is reliable and awesome.

2. Biological Replicates

Definition: Performing measurements on samples that come from *different biological sources*.

- **Example:** Take one blood sample from "some dude," one from "some other dude," and one from a third "other dude".
- These could be different people, different animals (mice), different plants, or different cell lines.

What do they tell you?

- **About the group:** They tell you about the average gene expression and variability within a *population* or *group* (e.g., "dudes in general").
- **Generalizability:** They are what allow you to **generalize your findings** to a broader group.

Important Caveat: Your generalization is limited by your samples. If you only sample "blue dudes," your results only tell a story about "blue dudes," not "orange dudes" or "ladies".

3. Summary: Which to Use?

- Do you want to talk about an **individual** or a **method**? → Use **Technical Replicates**.
- Do you want to talk about a **group** or **population**? → Use **Biological Replicates**.
- **Pro-Tip (for RNA-Seq):** It's almost always better to spend money on *more biological replicates* rather than technical replicates.

Video 2: Sample Size and Effective Sample Size

This video explains what " n " (sample size) *actually* means. **Spoiler:** It's the number of relevant replicates for your specific question.

1. What is Sample Size (n)?

Your sample size (n) is the number of **biological replicates**, *not* technical replicates.

- **Scenario 1:** You measure gene expression in **3 different blue dudes** (3 biological replicates).
 - Your hypothesis is about "blue dudes."
 - $n = 3$.
- **Scenario 2:** You measure the same **3 blue dudes**, but you measure *each of them twice* (3 biological replicates, 6 total measurements).
 - Your hypothesis is *still* about "blue dudes."
 - $n = 3$.
 - The extra measurements are technical replicates. They just make your measurement for each of the 3 dudes more precise, but they *do not* tell you more about the differences *between* dudes.
- **Exception:** If your hypothesis is about the *accuracy of your method*, then the technical replicates become your sample. If you ran 4 technical replicates, $n = 4$ for that specific question.

2. Effective Sample Size (The Complicated Part)

What happens when your biological replicates are not truly independent?

- **Scenario:** Your hypothesis is about "dudes in general."
 - You sample 1 blue dude, 1 orange dude, and 1 green dude ($n = 3$).
 - Then, you add the **identical twin** of the blue dude.
- **Question:** Is your sample size $n = 4$?
- **Answer:** No. It's somewhere **between 3 and 4**.
 - **Why?** The identical twins' measurements are not independent. Their genomes are the same, so their gene expression will be **highly correlated**.
 - Because they are correlated, the second twin provides *less new information* than a totally unrelated person would.

3. Calculating Effective Sample Size

When samples are correlated, they don't count as full individual samples. We must calculate an **effective sample size** (n_{eff}).

The video gives an example formula for a group of 2 related samples (the twins):

$$n_{eff} = \frac{n}{1 + (n - 1)\rho}$$

Where:

- n = number of correlated samples (e.g., 2 for the twins)
- ρ (rho) = the correlation between the samples

Example 1: High Correlation ($\rho = 0.7$)

The twins are very similar.

$$n_{eff} = \frac{2}{1 + (2 - 1)(0.7)} = \frac{2}{1.7} \approx 1.18$$

- These two highly-correlated twins only count as **1.18 people** in your study.

Example 2: Low Correlation ($\rho = 0.1$)

The twins are not very similar.

$$n_{eff} = \frac{2}{1 + (2 - 1)(0.1)} = \frac{2}{1.1} \approx 1.82$$

- These two weakly-correlated twins count as **1.82 people**.

Final Takeaway: Your true sample size (n) is the number of **independent biological units** relevant to your hypothesis. If your units are related (like twins, or samples from the same family), you must use the **effective sample size**.

http://googleusercontent.com/youtube_content/2 http://googleusercontent.com/youtube_content/3